Neural Wave 2026
PhDs but not for long

The R^3 ensamble

A robust, reliable reconstruction model for electricity demand

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Abstract

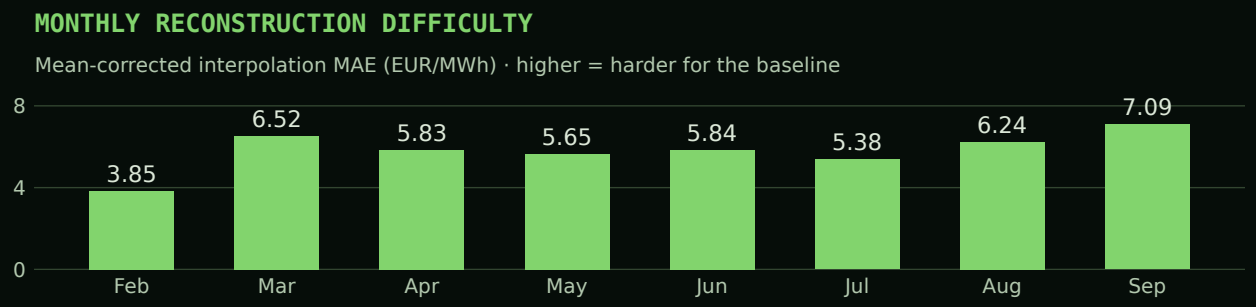
1. Summary of the project: Upsampling to quaterly data

This project's main task is upsampling electricity prices to quarterly hour resolution, that is, with a $4x$ oversampling ratio. For this task, a dataset of more than $130k$ entries is given, with data from four different countries from 1st of October of 2025 to 31 of September of 2026. This is a relative small dataset for this task, and the problem itself is slightly noisy, so robustness is a priority.

Our main goals in building a model can be summarized as 1) Precision and Robustness, 2) Coherent Validation and 3) Meaningful feature selection. In summary, we need a model with a strong performance in MAE terms, the challenge target metric, but that is also able to predict its own uncertainty. In order to not bias our research on an unusually hard test months, we also develop a more general model selection framework.

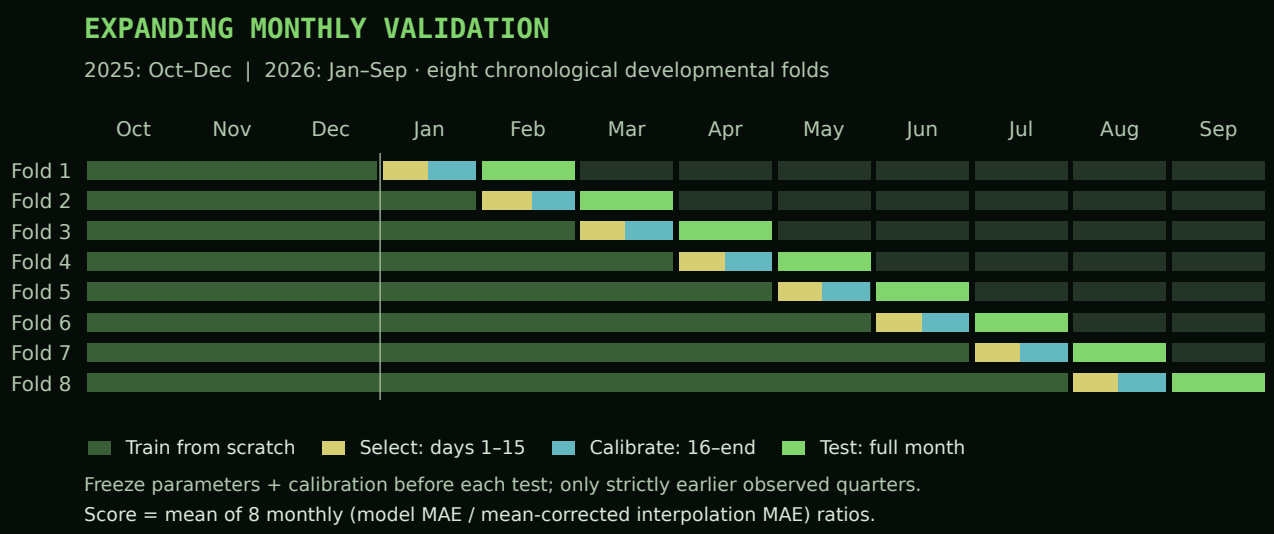
2. Validation

Time series models require a specific validation method that is different from other types of data, due to their temporal structure. Typically, the datasets are not randomly shuffled and then split as it is the case with images. Since we have to respect the temporal order, the usual approach consists in using the last part of the dataset as test, and the one before it as validation for hyperparameter tuning. However, in this small and noisy dataset this represents a problem: if we used the particularly noisy September 2026 to test the models, as shown in Figure 1, we could overfit our search on it. On top of that, the model might have encoded cyclic patterns and drifts that are incompatible with the test set. This also happens with the zone, but as we validated in all of them, this problem does not appear.



With this problem in mind, we apply an extending window cross-validation. The original cross validation is not applicable to this, as we would be using future time data in several folds. We respect the temporal order by splitting in month granularity, choosing a month from May to September, using that one as test, and the one before as validation. All other previous months will be used for training, and we will average the results of the several test months afterwards. This method is represented graphically in Figure 2.

Lastly, there are instances that might be by themselves really difficult to predict, as they are noisy and we do not have enough information to predict them. If we would select the model based on MAE only, we would be biasing ourselves into models that are good to predict smooth signals, as outliers will not be taken into account (in this case MSE could



be a better alternative. However, since we optimize for MAE, we forfeit this approach). In the model selection we aim to optimize $MAE/MAE_{interpolation}$ instead, that is, the precision the model adds to the linear interpolation baseline, so we can have a better idea of how good the model is on both smooth and difficult data.

3. Method

As stated beforehand, there is a concern about robustness, which provides justification for our choice of an ensemble of models. Individually, model predictions are easier to interpret, that is why we chose smaller deep learning models such as CNNs and MLPs in favor other architectures like maybe simpler but not as capable (e.g. Trees) or more complex and not interpretable (e.g. Transformers). This models are the following, with each one of them meant for a different feature scope:

- The MLPs, focus on the hourly high resolution data,
- The 1D CNNs, which we want to focus on intra zone relationships,
- The 2D CNNs, with the aim of aggregating inter zone data, applying kernels that can combine neightbour information due to relationships between the markets of each zone.

They are all training separately, and instead of predicting directly the curve, we predict the difference with respect to the linear interpolation, which gives us a good baseline. Additionally, instead of just predicting four values, one for each quarter, we predict three of them, and let the last missing information dimension be the mean, so we automatically and always fulfill the constrain of respecting the hourly mean. We do this by predicting values a, u and v. We perform the upsampling following $(a+u, a-u, -a+v, -a-v) + interpolation$. Then, when computing the mean, it will be equal to the interpolation one, making the constrain structural in our design. In Figure 3, top panel, we show a summary of the architecture. We combine the outputs of all the models to generate the final output.

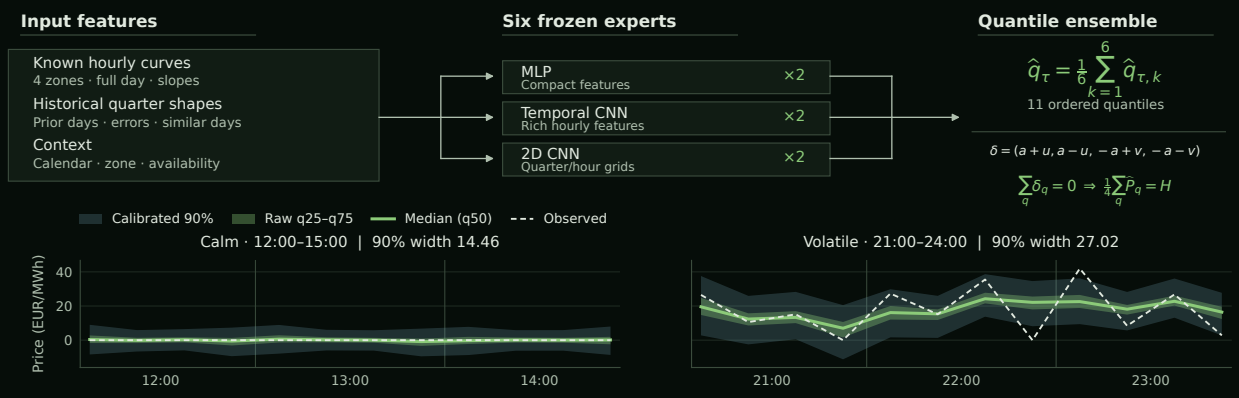
Lastly, we wanted to add some method of assessing the uncertainty of the model. For this matter, instead of just optimizing directly MAE, we optimize the pinball loss, for quantiles optimization. We define 11 quantiles, and use the median of it for the predictions. With this we will be able afterwards assess how confident is the prediction. An example of this is presented in Figure 3, down panel.

Regarding normalization in particular, we tested several types but found out that the classical mean/stdev normalization was working best. More complex techniques like RevIN seem to now work as well in this scenario.

4. Results and discussion

The resulting model obtains a 0.708 normalized MAE. This means that the model, with the validation structure shown, is a 29% better than the interpolation baseline. Additionally, the 90% quantile covers a 90.6% of cases correctly, which indicates a good calibration. The un-normalized MAE over the several months is $4.1 EUR/MWh$.

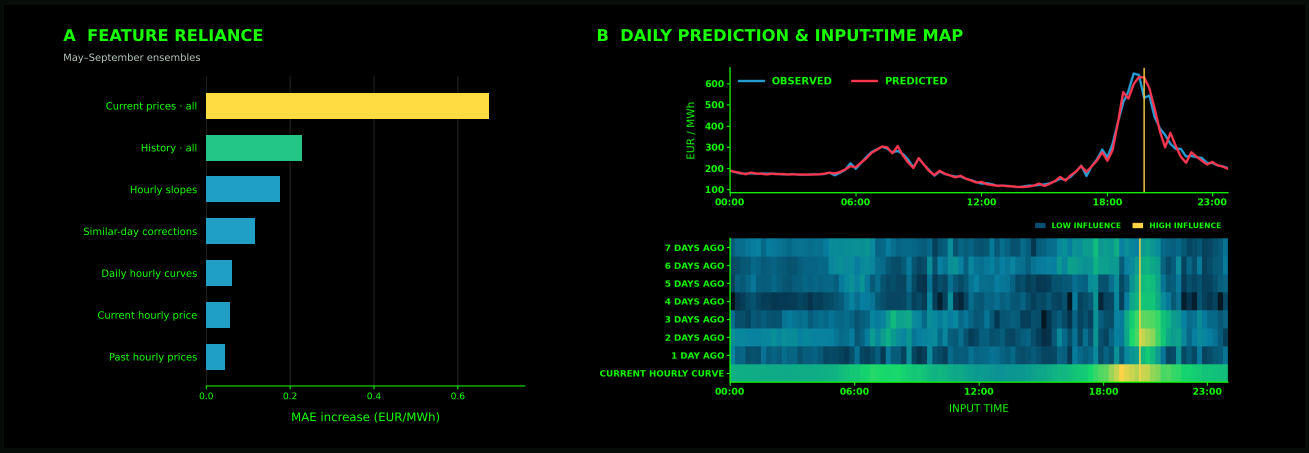
Result numbers typically do not explain the whole picture. For this reason, we conduct several interpretability analysis



aimed to shed more light into the model predictions, to look for possible fatal errors of the model or biases. In particular, we performed a feature reliance analysis for the ensemble and a GradCAM analysis aimed for the 1DCNNs submodules.

The first analysis, in Figure 4, shows that the most important features to the model turn out to be the most intuitive ones. Hourly prices for example, i.e. the down-sampled version, are the most important features. Other useful features turn out seasonality or the calendar.

In the second part of Figure 4 we can see the analysis focused on the 1DCNN. The models predictions align with our intuition: similar hours from the past days influence the model the most, while other parts (earlier or later) influence decreases. Also, the current day as shown in the feature importance, is more important than the lagged days. This input attribution experiment is performed by using a GradCAM variant.



5. Conclusion

With this project proposal we aim to showcase a model that is not only offers a decent performance at upsampling, but also one that one can trust due to its robustness and the coherence of predictions with the intuition of the users. We also provide an architectural constraint on the model output to satisfy the challenge requirements and propose a validation scheme that, joining the expanding window with the normalized MAE, for unbiased model selection.

We added more complementary information in this web extra docs.